DAVID KANG

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snnnue.github.io

INTERESTS

I am a fourth-year Life Sciences student at the University of Toronto and a research student in the Josselyn-Frankland Lab at SickKids. My research focuses on *memory engrams* and *fear memory circuits*, currently characterizing the temporal dynamics of Npas4 expression in the lateral amygdala. Building on this foundation, I am interested in *applying* computational and AI methods *to neuroscience and clinical research.*

EDUCATION

University of Toronto, *Toronto, ON* Sep 2023 - **Present**

- B.S. in Life Sciences, Victoria College
- Double Major in Physiology and Health & Disease

PROJECTS

[P1] **Characterizing the temporal dynamics of Npas4 expression in the lateral amygdala following fear memory encoding and retrieval**
David Kang
In Progress – UofT PSL498Y1 Project in Physiology 2026/27

[P2] **READYation – a proton flux forecasting system for astronaut radiation safety**
David Kang, Aaron Jang, Taryn Chong, Hee-won Jeong
Sheridan Datathon 2026, Google Developer Group on Sheridan College

RESEARCH EXPERIENCE

Josselyn Frankland Lab, SickKids PGCRL Toronto, Canada
Research student Sep 2026 – Present
Research volunteer Jan 2026 – Aug 2026

- Advisor: **Prof. Sheena Josselyn**
- Researching the temporal dynamics of Npas4 expression in the LA [P1].
- Performed fluorescence cell counting in Fiji/ImageJ.

Memory Biology Lab, KAIST Daejeon, Korea Jun 2025 – Jul 2025
Research student

- Advisor: **Prof. Jin Hee Han**
- Trained in stereotaxic viral surgery, perfusion, and brain slicing for engram research.

Dept. of Orthopaedics, Mount Sinai Hospital Toronto, Canada Nov 2024
Research volunteer

- Advisor: **Dr. Oleg Safir**
- Analyzed long-term post-operative outcomes for gluteus maximus patients using EMR.

WORK EXPERIENCE

East Asian Library, University of Toronto Toronto, Canada Sep 2025 – Aug 2026

- Supported library operations and patron services, assisting researchers navigating East Asian language collections and archival materials.

Wilson Heights Pharmacy Toronto, Canada Jun 2024 – Aug 2024
Pharmacy Assistant, Key Holder

Awards & Honors

Sheridan Datathon 2026 Winner, Google Developer Group on Sheridan College — 2026
- Placed first among 73 teams with READYation [P2] as first-time participants.

The Victoria Reunion Award, University of Toronto — 2026
- Awarded to students with overall A standing in Third Year; $1,000 scholarship.

Undergraduate Research Program, KAIST — 2025
- Selected for Biological Sciences URP; $600 research grant.

Volunteer Experience

Korean Language Teacher, Local Community Korean Language School — 2023 - Present

COVID-19 Safety & Screening Coordinator, Local Community Church — 2020 – 2022

Co-Op Clerical Assistant, Antenatal Postpartum High-Risk Unit, MSH — Feb – Nov 2022

Services & Leadership

Founder & President, Association for Research in Clinical Science, UofT — 2026 – Present

Academic Executive, Volunteer Organization of Korean Life Science Students, UofT — 2023 - 2026

Co-Director of Educational Events,
Association of Korean-Canadian Scientists & Engineers, YG UofT Chapter — 2024 - 2025